

A mechanistic approach on the solid state thermal decomposition of bimetallic oxalate coordination compounds of Mn(II), Fe(II) and Cu(II) with cobalt

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Abstract

The mixed metal carboxylates, manganese(II)bis(oxalato)cobaltate(II)hydrate (MOC), iron(II)bis(oxalato)cobaltate(II)pentahydrate (IOC), and copper(II)bis(oxalato)cobaltate(II)trihydrate (COC) have been synthesized and characterized by elemental analysis, IR spectral and X-ray powder diffraction studies. Thermal decomposition studies (TG, DTG and DTA) in air showed that the compound MOC decomposed mainly to MnCoO_3 at 315 °C through the formation of a complete anhydrous compound. The species CoFeO_3 is generated at 274 °C from the compound IOC through the formation of a very transient anhydrous compound at around 218 °C. In case of the compound, COC the end product was found to be CuCoO_2 at ~1000 °C through the existence of a quite stable mixture of CuO and CoO at ~318 °C. DSC studies in nitrogen indicated the formation of a mixture of MnO and CoO in case of MOC, cobalt ferrite in case of IOC and a brown mixture of Cu_2O and CoO from COC as end products. The kinetic parameters have been evaluated for both the dehydration and decomposition steps of all the three compounds using four non-mechanistic equations. Using seven mechanistic equations, the rate controlling processes of the dehydration and decomposition mechanism are also inferred. The kinetic parameters, ΔH and ΔS obtained from DSC are presented. IR and X-ray powder diffraction studies identified some of the decomposition products.

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1. Introduction

In the last two decades, the spinel ferrites which is always technologically important to the microwave industries, has been important for high speed digital tapes or disk recording [1]. It has future application for use in levitated railway systems [2] as repulsive suspension. This has tempted several workers to prepare the ultrafine ferrite particles [3–5]. The synthesis of mixed metal oxide is possible through the thermal decomposition of their carboxylate compounds. The formation of ultrafine NiFe_2O_4 and CoFe_2O_4 were reported through multi-step decomposition reaction from respective bimetallic nickel iron citrate precursor, $\text{Ni}_3\text{Fe}_6\text{O}_4(\text{C}_6\text{H}_6\text{O}_7)_8 \cdot 6\text{H}_2\text{O}$ [6] and cobalt iron citrate precursor, $\text{Co}_3\text{Fe}_6\text{O}_4(\text{C}_6\text{H}_6\text{O}_7)_8 \cdot 6\text{H}_2\text{O}$ [7]. Thermal

decomposition of strontium iron citrate precursor, $\text{SrFe}_{12}\text{O}_6(-\text{C}_6\text{H}_6\text{O}_7)_{13}$ [8], the residue of single-phase amorphous strontium ferrite, $\text{SrFe}_{12}\text{O}_{19}$ generated below 550 °C, which crystallizes completely above this temperature. Fine particles of lithium ferrite, LiFeO_2 at 490 °C and cesium ferrite, CsFeO_2 at 700 °C are generated in air through a solid state reaction between the carbonates of lithium or cesium with $\alpha\text{-Fe}_2\text{O}_3$, which were formed as intermediates in the decomposition of respective lithium iron malonate, $\text{Li}_3[\text{Fe}(\text{CH}_2\text{C}_2\text{O}_4)_3] \cdot 4\text{H}_2\text{O}$ [9] and cesium iron oxalate, $\text{Cs}_3\text{Fe}(\text{C}_2\text{O}_4)_3 \cdot 2\text{H}_2\text{O}$ [10] precursors. A solid state reaction between MgO and $\alpha\text{-Fe}_2\text{O}_3$ to yield magnesium ferrite MgFe_2O_4 at 600 °C, a temperature lower than for ceramic method was reported [11] from magnesium tris(maleato)ferrate(II)dodecahydrate, $\text{Mg}_3[\text{Fe}(\text{C}_2\text{H}_2\text{C}_2\text{O}_4)_3]_2 \cdot 12\text{H}_2\text{O}$. In the last decade, the pollution by diesel-engine exhausts has become more and more serious. NO_x and soot are main components to be removed from diesel exhausts. Among bimetallic spinel-type oxides, CuFe_2O_4 a useful catalyst which shows intermediate

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activity and exceptionally high and low selectivity, respectively, to N_2 and N_2O formation [12]. A mixture of tetragonal $CuFe_2O_4$, $\alpha-Fe_2O_3$ and CuO is found to be generated from the thermolysis of $(NH_4)_8[Fe_2Cu(C_2O_4)_8]$ and a clean tetragonal $CuFe_2O_4$ was obtained from $[Fe_2Cu(C_2O_4)_2(OH)_4] \cdot 4H_2O$ in the study of polynuclear coordination compounds [13]. From a technological viewpoint, rare earth orthoferrites $RFeO_3$ ($R = Y$ or lanthanide) have been proposed for use in resonance-based devices such as logic and memory elements, and as lasers and light modulators in optical applications [14,15].

The catalytic activity of the mixed oxide obtained [16] from the coprecipitated oxalates of $Co(II)$ and $Cu(II)$ was found to be lower than that of the oxides obtained from the mechanical mixture at the same temperature. In recent past, copper ferrite was reported [17] to be generated from the thermal decomposition of coordination compounds containing copper, iron and oxalic anion as ligand. The effect of calcination temperature on the physico-morphological properties of $CoC_2O_4 \cdot 2H_2O-FeC_2O_4 \cdot 2H_2O$ mixture has been investigated [18] in detail, where decomposition proceeds via four well-defined steps leading to $CoFe_2O_4$ as end product beyond $920^\circ C$. A single-phase manganese ferrite powder, $MnFe_2O_4$ with a cubic structure also been synthesized [19] at $1100^\circ C$ through the thermal decomposition reaction of $MnC_2O_4 \cdot 2H_2O-FeC_2O_4 \cdot 2H_2O$ mixture in air. Very recently, the effect of varying amount of carbon support on the decomposition of mixed Mn and Co nitrates and the formation of spinel $MnCo_2O_4$ was reported [20]. It was found that both the decomposition of nitrates and the presence of spinel oxide enhanced the oxidation of carbon. The pyrolysis of polymetallic oxalate precursor powder, $Ba_{1-x}Sr_xTiO(C_2O_4)_2 \cdot 4H_2O$ ($x = 0.25$) at $730^\circ C/4$ h in air reported [21] to produce $Ba_{1-x}Sr_xTiO_3$ powders, which was found to be highly pure, cubic, stoichiometric and sub-micron in size. Thermal decomposition of bimetallic oxalate of iron, $Fe[Fe(C_2O_4)_3] \cdot 4H_2O$ is also reported [22].

In the studies reported above, during the extraction of the oxide end products the precursors undergoes several dehydration and decomposition reactions and it is worthwhile to explore the mechanism and kinetics of the various steps. In continuation with our ongoing efforts [23–26] in the field of mixed metal carboxylates decomposition, herein we report the synthesis, characterization and the thermal decomposition of $Mn[Co(C_2O_4)_2] \cdot 3.5H_2O$, $Fe[Co(C_2O_4)_2] \cdot 5H_2O$ and $Cu[Co(C_2O_4)_2] \cdot 3H_2O$. Emphasis was given on mechanistic approach on the dehydration and decomposition steps of all the complexes.

2. Experimental

2.1. Preparation of the complexes

The compounds were prepared from cobalt(II) chloride by the similar methods adopted earlier [24]. All the reagents used were AR grade. The chloride free cobalt hydroxide prepared from cobalt chloride was dissolved in glacial acetic acid and heated on a steam bath. Adding manganese chloride followed

by dropwise addition of saturated solution of oxalic acid separated a light pink precipitate. Similarly, after adding ferrous chloride and cuprous chloride, the respective precipitates of light yellowish pink and bluish pink colour were isolated. The precipitates were washed several times with distilled water and stored in a desiccator over calcium chloride. The water contents of the compounds were determined gravimetrically and metal contents were estimated using standard methods [27].

2.2. Instrumentation

Elemental analyses were done with a Carlo Erba 1108 elemental analyzer. IR ($200-4000\text{ cm}^{-1}$) spectra of the compounds were recorded on a Perkin-Elmer IR 883 spectrophotometer. TG, DTG and separately DTA were carried out at $10^\circ C/min$ in static air with a Shimadzu DT-30B thermal analyzer. The amount of sample taken was 7.08 mg in TG and 10.0 mg in DTA for the compound MOC, 7.53 mg in TG and 20 mg in DTA in case of IOC, and for the compound COC the respective amounts were 8.13 and 10 mg. DSC curves were recorded with a TA instrument DSC-2010 at a heating rate of $10^\circ C/min$ in nitrogen atmosphere with a flow rate of 50 mL/min up to $550^\circ C$ for the compound MOC and COC, whereas scanning was done up to $600^\circ C$ for IOC. The sample mass taken for the respective compounds, MOC, IOC and COC were 5.90, 5.53 and 6.81 mg. The instrument was calibrated with indium (purity 99.9%). The enthalpy (ΔH) values are obtained after integration of the peaks through universal V2.3C TA software in J/g and converted into kJ/mol using molecular mass, and entropies (ΔS) are calculated using simple well-known thermodynamic relation incorporating the temperature of respective peak maxima. The powder X-ray diffraction (XRD) pattern of the compounds and some of the decomposition products were taken at ambient temperature using a Geigerflex Microprocessor controlled Automated Rigaku (Japan) X-ray diffractometer system D/Max IIC with SASG accessory. The Ni filter with monochromatic $Cu\ K\alpha$ ($\lambda = 1.5406\text{ \AA}$) radiation at 30 kV and 20 mA in the wide angle $5^\circ < 2\theta < 160^\circ$ were used for the XRD.

2.3. The kinetics of decomposition

Kinetic parameters, such as, activation energy (E^*), pre-exponential factor (A) and the order of reaction (n) of dehydration and decomposition steps were calculated from non-isothermal TG curves using different non-mechanistic equations, such as, Freeman and Carroll [28], Flynn and Wall [29], Coats and Redfern [30], and MacCallum and Tanner [31].

Several authors [32–35] have discussed evaluation of the mechanism of reaction using mechanistic equations. We have used the following kinetic expression [35]:

$$\frac{\Delta \ln \alpha'}{\Delta \ln(1-\alpha)} = -\frac{E^*}{R} \frac{\Delta(1/T)}{\Delta \ln(1-\alpha)} + \frac{\Delta \ln f(\alpha)}{\Delta \ln(1-\alpha)}, \quad (1)$$

Table 1

Selected bands in the IR spectra of the complexes, $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$, $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ and $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$

IR bands (cm^{-1})			Assignment
$\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$	$\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$	$\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$	
3000–4000 bS	2920–3920 bS	3040–3800 bS	ν_{sy} (O–H) + ν_{asy} (O–H) or hydrogen bonding
1650 S	1620 S	1640 S	δ_{sy} (H–O–H)
1440–1900 bS	1390–1780 bS	1480–1800 bS	ν_{asy} (C=O)
1380 mS	1360 m	1370 mS	ν_{sy} (C–O) and/or ν (C–C)
1340 S	1320 mS	1320 mS	ν_{sy} (C–O) and/or δ (O–C=O)
840 S, 750 mS	820 m	820 m	ν (M–O) and/or δ (O–C=O)
630 mS	610 s	620 s	lattice water
480 S (550 sh)	460 S	470 mS	ν (M–O) and/or ring deformation
330 s	340 s	360 m	δ (O–C=O) and/or ν (C–C)
270 s	280 s	300 m	π (out-of-plane bending)

(b), broad; (m), medium; (S), strong; (s), small; (vs), very small.

where the terms have their usual meaning and $f(\alpha)$ is a function depending on the actual mechanism of the process. A series of seven forms of $f(\alpha)$ are proposed [35] out of the possible mechanisms [33] of thermal decomposition. Thus, a plot of $\Delta \ln \alpha' - \Delta \ln f(\alpha) / \Delta \ln(1 - \alpha)$ against $\Delta(1/T) / \Delta \ln(1 - \alpha)$ is a straight line whose slope is $-E^*/R$, irrespective of the form of $f(\alpha)$ employed. One can select the $f(\alpha)$ that best fits the actual mechanism of reaction corresponding to the intercept value and correlation coefficient close to zero and unity, respectively. The actual mechanisms are as follows:

Mechanism	Rate-controlling process
D1	One-dimensional diffusion
D2	Two-dimensional diffusion
D3	Three-dimensional diffusion; Jander equation
D4	Three-dimensional diffusion; Ginstling–Brounshtein equation
F1	Random nucleation
R2	Phase-boundary reaction (cylindrical symmetry)
R3	Phase-boundary reaction (spherical symmetry)

Further, to test the correctness and validity of the above conclusion regarding the identification of the actual mechanism of the process, the Arrhenius equation of the following type is used:

$$\ln \alpha' - \ln f(\alpha) = \ln \left(\frac{A}{\beta} \right) - \frac{E^*}{RT} \quad (2)$$

where the terms have their usual meaning. The slope, $-E^*/R$, and intercept, $\ln(A/\beta)$, can be obtained from the plot of $\ln \alpha' - \ln f(\alpha)$ against $1/T$ which is a straight line. It now follows that the mechanism proposed on the basis of the kinetic equation (1) is correct provided the E^* value obtained from the above plot of (2) turns out to be same.

Thus, we have evaluated the E^* from TG curves employing the kinetic Eqs. (1) and (2) for seven forms of $f(\alpha)$ and thus inferred the rate controlling process for the dehydration and decomposition steps of all the three complexes as described above. The details of the kinetic parameters calculated are available as supplementary material in the online publication of the paper.

3. Results and discussion

The light pink compound, $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$ (MOC), light yellowish pink compound, $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ (IOC) and the bluish pink compound, $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$ (COC) were insoluble in water or in common organic solvents. However, they decomposed in the presence of strong acid or alkali. The analytical and microanalytical results were consistent with the

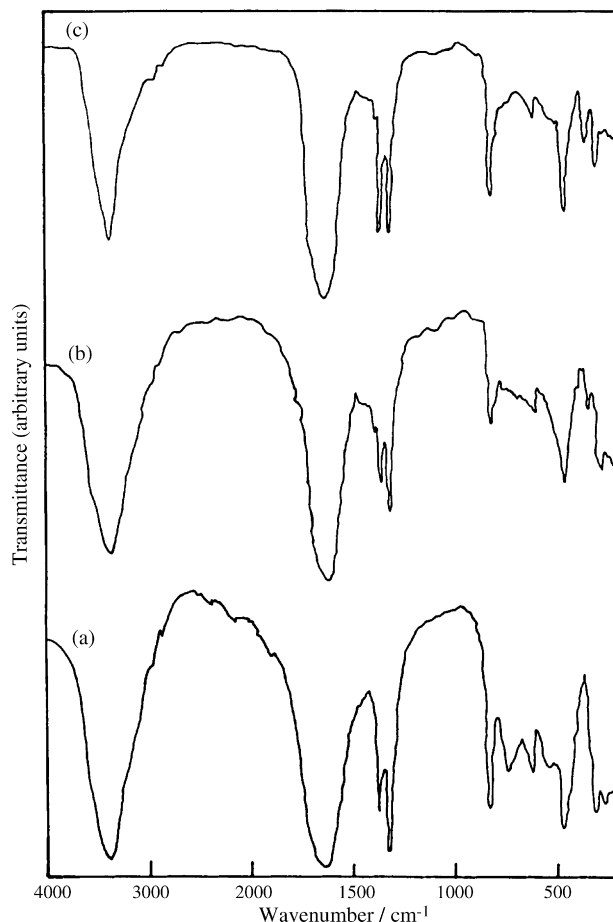


Fig. 1. IR spectra of the complexes, (a) $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$, (b) $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ and (c) $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$.

proposed formula of the compounds. Analysis as calculated for $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$: Mn, 15.56%; Co, 16.69%; C, 13.61%; H, 1.99% and H_2O , 17.86%; found: Mn, 15.02%; Co, 15.94%; C, 12.54%; H, 2.10% and H_2O , 16.90%; calculated for $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$: Fe, 14.66%; Co, 15.47%; C, 12.60%; H, 2.65% and H_2O , 23.65%; found: Fe, 14.23%; Co, 15.00%; C, 12.42%; H, 2.60% and H_2O , 22.65%; calculated for $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$: Cu, 18.02%; Co, 16.72%; C, 13.63%; H, 1.72% and H_2O , 15.32%; found: Cu, 18.67%; Co, 15.98%; C, 13.54%; H, 2.00% and H_2O , 15.18%. The chelating character of the oxalato groups as well as the presence of lattice water has been concluded on the basis of earlier studies [24,36]. The assigned IR peaks were shown in Table 1 and Fig. 1. Powder XRD pattern (Table 2) of all the three compounds differed from one other and suggested that they are not isomorphous and are crystalline in nature.

The thermal profiles (TG, DTG and DTA) of $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$ are shown in Fig. 2. In the TG curve a steep slope between 142 and 200 °C with a mass loss of 18% (calcd., 17.86%) indicates the removal of all the three and half molecules of water. A DTG change between 140 and 220 °C, and an endothermic peak in the range 140–222 °C ($\Delta T_{\text{min}} = 180$ °C) in DTA corresponds to the dehydration step. The anhydrous compound is stable up to 280 °C in TG, the precursor is heated up to 230 °C in furnace and the pyrolysed product is subjected to powder XRD and IR studies.

Differences in the d -spacing in the XRD data with the precursor are due to change in the internal structure on removal of molecules of water. However, a few d -values are similar to d -spacing of the precursor, which indicates the retention of chelated oxalato groups. The IR bands also confirmed the retention of $\text{C}_2\text{O}_4^{2-}$ groups. The kinetic parameters (activation energy (E^*), pre-exponential factor (A), and order of reaction (n)) calculated for the dehydration steps from TG slope using four non-mechanistic equations [28–31] are included in Table 4. Slight differences in the values of E^* for the same step using different methods are observed like obtained by many workers [24,25,37,38]. The variations may be due to the different methods based on different approaches and assumptions, and so those have different inherent limitations in the methods of calculation. The E^* of the step was calculated using seven mechanistic equations [32–35] and the rate controlling process of the dehydration step is found to be two-dimensional diffusion (D2). The mechanistic study suggested that the dehydration had taken place with higher E^* values in most of the mechanisms, whereas lower values are calculated adopting non-mechanistic equations. The deaquated compound, which is stable up to 280 °C, decomposes within a short temperature range as shown by a steep slope in TG in the range 280–315 °C with a 51% mass loss. An exotherm between 267 and 320 °C with $\Delta T_{\text{max}} = 280$ °C in DTA and a distinct change in the range 280–363 °C in DTG are accounted for the decomposition step.

Table 2

Prominent lines in the X-ray powder diffraction pattern of the complexes, $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$ (MOC), $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ (IOC), $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$ (COC) and pyrolysed products of MOC at 340 °C, IOC at 300 °C, and COC at 330° and 1000 °C

MOC		IOC		COC		Pyrolysed product of MOC at 340 °C		Pyrolysed product of IOC at 300 °C		Pyrolysed product of COC at 330 °C		Pyrolysed product of COC at 1000 °C	
$d(\text{\AA})$	I(Rel)	$d(\text{\AA})$	I(Rel)	$d(\text{\AA})$	I(Rel)	$d(\text{\AA})$	I(Rel)	$d(\text{\AA})$	I(Rel)	$d(\text{\AA})$	I(Rel)	$d(\text{\AA})$	I(Rel)
7.06	68	11.37	90	4.84	100	3.63	42	3.74	44	2.52	100	2.80	85
4.86	100	6.25	77	3.94	90	3.20	75	2.95	40	2.44	90	2.42	43
3.95	67	4.96	100	3.02	76	2.70	100	2.87	61	2.33	84	2.34	100
3.05	72	3.14	64	2.49	67	2.45	70	2.82	37	2.31	22	2.28	60
1.94	47	2.38	50	1.94	69	2.27	19	2.70	100	2.28	33	2.25	41
1.66	42	2.03	48	1.79	68	2.13	28	2.66	32	2.12	68	2.11	33
1.51	39	1.61	42	1.59	64	1.97	16	2.51	52	1.98	36	2.06	20
1.03	29	1.39	36	1.49	65	1.82	22	2.41	11	1.88	44	1.98	36
0.93	26	1.25	37	1.27	61	1.78	80	2.33	70	1.78	39	1.86	27
0.89	29	1.11	33	1.11	60	1.67	55	2.28	18	1.62	31	1.72	30
0.87	25	1.05	34	0.98	59	1.60	32	2.21	33	1.58	13	1.49	24
0.86	27	0.91	31	0.91	57	1.57	30	1.98	16	1.52	53	1.38	20
0.80	24	0.86	31	0.81	58	1.48	28	1.88	14	1.41	20	1.27	39
0.79	26	0.82	34			1.45	20	1.84	22	1.37	12	1.20	46
		0.80	33			1.39	42	1.78	36	1.26	11	1.18	29
		0.79	34			1.31	35	1.72	54	1.18	36	1.17	43
						1.06	18	1.69	24	0.99	18	1.03	20
								1.64	41	0.92	24	1.00	28
								1.60	28	0.86	30	0.98	37
								1.48	19	0.82	14	0.97	28
								1.37	20			0.86	39
								1.32	9				
								1.26	17				
								1.18	40				
								0.99	7				
								0.85	25				

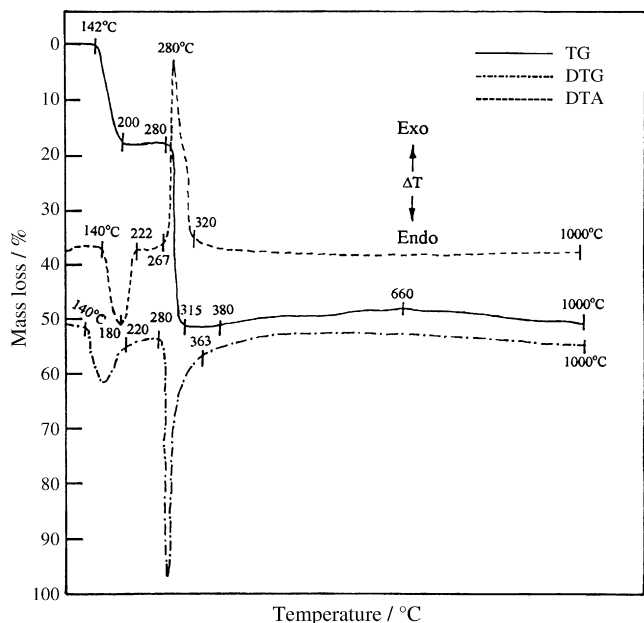


Fig. 2. TG, DTG and DTA curves of $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$ in air at $10^\circ\text{C}/\text{min}$.

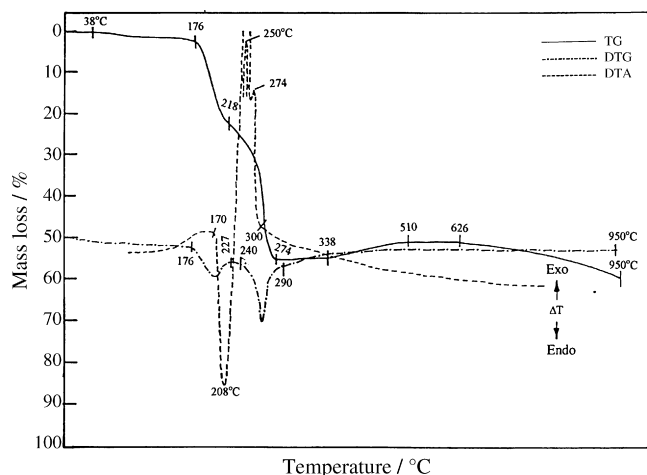


Fig. 3. TG, DTG and DTA curves of $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ in air at $10^\circ\text{C}/\text{min}$.

The species formed at 315°C in TG with 51% mass loss apparently indicates the formation of a mixed metal oxide $\text{CoMnO}_{3.5}$ (calcd. mass loss, 51.87%) which may be an intimate mixture of MnO_2 and $1/2\text{Co}_2\text{O}_3$. As the species formed at 315°C was stable up to 380°C in TG the XRD pattern (Table 2) of the black product isolated after independent pyrolysis up to 340°C indicates [39] the presence of CoMnO_3 . The calculated mass loss to get CoMnO_3 is 54.14%. A few d -values in XRD data are similar to the value of MnO_2 and Co_2O_3 which indicates their presence in trace, and which in turn shows the difference in weight loss. The IR peaks at 650, 615, 560, 480, 400 and 330 cm^{-1} of the species further confirmed [41] their presence. The E^* value for the decomposition step calculated using four non-mechanistic methods are found (Table 4) to be much higher than the dehydration step. The kinetic analysis of the TG data for E^* values using mechanistic equations showed dominance of kinetic control mechanism based on phase-boundary reaction having cylindrical symmetry (R2). All of the E^* values calculated using mechanistic equations for the decomposition step are found to be much

higher than the values calculated (Table 4) using non-mechanistic equations. The mass gain of 3% in the range $380\text{--}660^\circ\text{C}$ may be due to the slow formation of carbide [24–26] through the solid–gas reaction of evolved carbon monoxide in contact with the solid products, which led disproportionation of $\text{CO}(\text{g})$ to C and $\text{CO}_2(\text{g})$. The free unoxidised and unreacted carbon may also increase the mass of residue [23,25]. Finally, the mass loss of 51% in TG at 1000°C may be due to the formation of a residue similar to the species formed at 315°C and may be contaminated with traces of oxides and carbides of manganese and cobalt. The DSC profile (Table 3) in nitrogen shows two distinct endothermic peaks. The first peak is for dehydration. The second endothermic peak is for decomposition of the deaquated compound. The change of enthalpy (ΔH) and entropy (ΔS) for the steps are included in Table 3. Further, any changes are not recorded in the profile up to 550°C . The mass loss of 40.68% after scanning up to 550°C with the generation of a black residue which may be a mixture of 2MnO and CoO .

In thermal study of $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ (IOC) in air (Fig. 3) the mass loss started slowly from 38°C in TG curve and an inclined slope from 176 to 218°C with a mass loss of 22.5% (calcd., 23.65%) indicates the removal of all the five molecules of water. Initially the flat slope up to 176°C with a 2.5% mass loss suggested the release of half molecule of water. The kinetic

Table 3
DSC data of $\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$, $\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$ and $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$ in nitrogen at $10^\circ\text{C}/\text{min}$

Compound	Step	Temperature range ($^\circ\text{C}$)	Peak temperature ($^\circ\text{C}$)	$\log Z$ (min^{-1})	E^* (kJ mol^{-1})	Order of reaction	ΔH (kJ mol^{-1})	ΔS ($\text{JK}^{-1} \text{mol}^{-1}$)
$\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$	1	146–217 (endo.)	185.44				213.97	466.73
	2	340–460 (endo.)	424.38				220.97	316.38
$\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$	1	62–126 (endo.)	85.98				7.50	20.90
	2	170–257 (endo.)	216.24				211.51	432.32
	3	340–435 (endo.)	391.63				126.79	190.78
$\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$	1	150–255 (endo.)	212.35				80.45	165.77
	2	290–342 (exo.)	308.16	47.31	526.80	1.05	17.38	29.90
	3	360–424 (endo.)	390.57				44.11	66.47

Table 4

Kinetic data evaluated by the methods (a) Freeman and Carroll, (b) Flynn and Wall, (c) Coats and Redfern, and (d) MacCallum and Tanner equation

Compound	Step	Method	$E^*/(\text{kJ mol}^{-1})$	Order of reaction (n)	A/s^{-1}	Reaction
$\text{Mn}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3.5\text{H}_2\text{O}$	1	a	12.56	0.5	7.6×10^4 2.0×10^5	Dehydration
		b	13.10	0.8		
		c	10.00			
		d	13.62			
	2	a	92.32	1.0	8.2×10^9 1.6×10^8	Decomposition
		b	89.10	0.5		
		c	95.00			
		d	90.86			
$\text{Fe}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 5\text{H}_2\text{O}$	1	a	22.00	0.5	8.2×10^7 2.5×10^6	Dehydration
		b	21.86	0.5		
		c	26.12			
		d	20.36			
	2	a	112.86	1.0	3.2×10^{12} 3.8×10^{14}	Decomposition
		b	110.32	0.8		
		c	109.52			
		d	102.10			
$\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$	1	a	18.00	0.5	1.2×10^3 5.0×10^5	Dehydration
		b	15.52	0.5		
		c	13.12			
		d	17.63			
	2	a	122.00	1.0	3.2×10^9 5.6×10^7	Decomposition
		b	124.32	0.5		
		c	121.86			
		d	118.35			

parameters (activation energy (E^*), pre-exponential factor (A), and order of reaction (n)) calculated for the dehydration steps from TG slope using four non-mechanistic equations [28–31] are included in Table 4. Slight differences in the values of E^* for the same step using different methods are observed like obtained by many workers [25,37,38,40]. The E^* of the step were calculated using seven mechanistic equations [32–35] and the rate controlling process of the dehydration step is found to be two-dimensional diffusion (D2). The mechanistic study suggested that the dehydration had taken place with higher E^* values in most of the mechanisms, whereas lower values are calculated adopting non-mechanistic equations. The mass loss continues in TG and showing another inclined slope from 218 to 274 °C with 55.5% mass loss, which apparently indicates the formation of a mixed metal oxide CoFeO_3 that may be an intimate mixture of the species, $1/2\text{Co}_2\text{O}_3$ and $1/2\text{Fe}_2\text{O}_3$. The calculated mass loss to get the mixture is 57.26%. As the species is stable up to 338 °C in TG, the XRD of the pyrolysed products at ~ 300 °C confirmed the presence of Co_2O_3 and Fe_2O_3 , and a trace of CoFeO_3 . Further, the IR peaks at 560, 500, 465, 410, 350 and 325 cm^{-1} give credence to the presence of the oxides of cobalt and iron. In DTG two subsequent changes between 176 and 227 °C and 240–290 °C corresponds to these dehydration and decomposition steps. In DTA curve the peak in the range of 170° to 300 °C which consists of an endotherm with ΔT_{min} of 208 °C and an exotherm with ΔT_{max} of 250 °C corresponds to the dehydration and decomposition phenomena. The E^* value for the decomposition step calculated using four non-mechanistic methods are found (Table 4) to be much higher than the dehydration step. The kinetic analysis of the TG data

for E^* values using mechanistic equations showed dominance of kinetic control mechanism based on random nucleation. All of the E^* values calculated using mechanistic equations for the decomposition step are found to be much higher than the values calculated (Table 4) using non-mechanistic equations. The TG curve is almost stable up to 338 °C and then a mass gain of 4% up to 510 °C is observed (obsd. mass loss, 51.5%) which is almost stable up to 626 °C (obsd. mass loss, 51.8%). The gain in mass may be due to the slow formation of carbide [25] like observed in manganese cobalt complex. The free unoxidised and unreacted carbon may also increase the mass of residue [23,25]. The generation of several intermediates which consists also carbides and solid carbon and consequent weight gain of similar nature was very recently reported [26] in our manganese–nickel oxalato coordination compound. The species generated at 626 °C starts losing mass and at the end of the scanning up to 950 °C with a mass loss of 61% attributed the formation of a mixture of a mixed metal oxide CoFeO_2 (calcd. mass loss, 61.46%) which is an intimate mixture of CoO and FeO . The DSC data (Table 3) in nitrogen shows one flat and two distinct endothermic peaks. The first flat peak is for the removal of first molecule of water. The second sharp endothermic peak is for the release of remaining four molecules of water accompanied by the partial decomposition of the deaquated compound. Further, another endotherm in the range between 340 and 435 °C is due to further decomposition of the intermediate species formed after second endotherm. The ΔH and ΔS for all the steps can be seen in Table 3. The mass loss of 38.16% after scanning up to 600 °C with the generation of a black residue of cobalt ferrite, CoFe_2O_4 , which may be

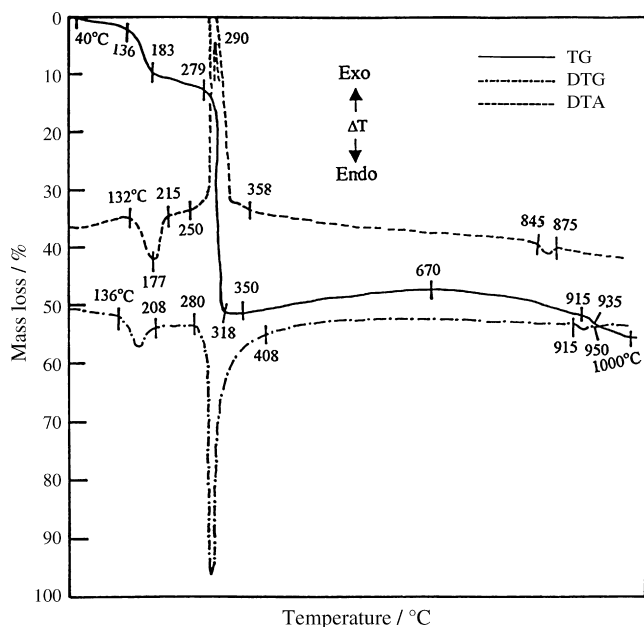


Fig. 4. TG, DTG and DTA curves of $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$ in air at $10^\circ\text{C}/\text{min}$.

generated through solid–solid reaction of the species CoO and Fe_2O_3 which are possibly formed as intermediates in the decomposition process.

The thermal profile (TG, DTG and DTA) of $\text{Cu}[\text{Co}(\text{C}_2\text{O}_4)_2] \cdot 3\text{H}_2\text{O}$ are shown in Fig. 4. The mass loss starts at 40°C in TG and an inclined slope up to 183°C with a mass loss of 9.5% indicates the removal of two molecules of water. A change between 136 and 208°C in DTG and an endothermic peak in the range 132 – 215°C ($\Delta T_{\text{min}} = 177^\circ\text{C}$) corresponds to the release of water molecules. The kinetic parameters (activation energy (E^*), pre-exponential factor (A), and order of reaction (n)) calculated for the step from TG slope using four non-mechanistic equations [28–31] are included in Table 4. The E^* of the step was calculated using seven mechanistic equations [32–35] and the rate controlling process is found to be one-dimensional diffusion. The mass loss continues in TG with a break at 279°C and then a very steep slope up to 318°C with a mass loss of 51% is for the decomposition of the compound accompanied by the release of last water molecule. The product at 318°C apparently indicates it to be a mixture of CuO and CoO . The calculated mass loss to get this mixture species is 56.18%. The difference in mass loss indicates the product is contaminated with other species. The mixture species is stable up to 350°C in TG which further gained mass of 3.5% up to 670°C indicates the gradual oxidation of species formed at 318°C . The DTG change in the range 280 – 408°C and an exothermic peak between 250 and 358°C ($\Delta T_{\text{max}} = 290^\circ\text{C}$) is for the decomposition of the compound. The XRD data (Table 2) of the pyrolysed species isolated at $\sim 330^\circ\text{C}$ confirmed [39] the presence of CuO and CoO . The IR bands at 610 , 495 , 460 , 408 and $\sim 350\text{ cm}^{-1}$ further substantiates their presence in the species. The E^* value for the major decomposition step calculated using four non-mechanistic methods are found (Table 4) to be much higher than the

dehydration step. The kinetic analysis of the TG data for E^* values using mechanistic equations showed dominance of kinetic control mechanism based on phase-boundary reaction having spherical symmetry (R3). In TG from 670°C further continuous slow decrease of mass with a break at 935°C with a mass loss of 53.5% indicates the formation of a mixture of CuO and Co_2O_3 (calcd., 53.92%). A small endothermic peak between 845 and 875°C in DTA and a change in the range 915 – 950°C in DTG is due to the changes take place in the intermediates. Finally, at 1000°C in TG with a mass loss of 56% is due to the formation of CuCoO_2 (calcd., 56.18%). The XRD data (Table 2) of the species pyrolysed at 1000°C confirmed [39] it to be CuCoO_2 . The DSC profile in nitrogen up to 550°C can be seen in Table 3. The first sharp endothermic peak correspond to the release of all water molecules accompanied by the partial decomposition of anhydrous species. The intermediate species was further decomposed as evidence by an exothermic peak in the range 290 – 342°C followed by subsequent decomposition corresponding to another endothermic peak between 360 and 424°C . The corresponding kinetic parameters for all the steps are shown in Table 3. The exothermic decomposition step was found to be first order with E^* value of 526.8 kJmol^{-1} which is much higher than the value calculated from the TG curve. Moreover, the exothermic decomposition was not observed in DSC study of manganese and iron compounds. The product formed at 424°C is stable in nitrogen indicated by no further change of DSC profile. The mass loss of 39.35% of the annealed sample up to 550°C is attributed to the formation of a brown mixture species of Cu_2O and CoO as an end product (calcd. mass loss, 38.16%).

The water vapour, carbon monoxide and carbon dioxide evolved during decomposition of all the three complexes were identified by IR spectroscopy [22,36].

4. Conclusions

Thermal decomposition in air of the three complexes, MOC, IOC and COC the generation of mixed metal oxides of the type $\text{M}_1\text{M}_2\text{O}_x$ were observed as main decomposition products. The result of thermal analysis in air does not show substantial differences in decomposition products in all the three complexes. However, the DSC study in nitrogen revealed the formation of a mixture of MnO and CoO at 550°C in case of MOC, CoFe_2O_4 at 600°C from IOC and a mixture of Cu_2O and CoO at 550°C in case of COC. Copper cobalt oxalato complex (COC) undergoes intermittent exothermic decomposition in DSC study in nitrogen, which is not observed in manganese and iron complexes. Kinetic analysis by advanced mechanistic approach on the solid state thermal dehydration enabled us to determine the rate controlling kinetic model as two-dimensional diffusion for the complexes MOC and IOC, and one-dimensional diffusion for the complex COC. The dominating role of phase-boundary-controlled reactions having cylindrical symmetry for decomposition (in case of manganese cobalt complex, MOC) and spherical symmetry are observed in the decomposition of copper cobalt complex (COC). However in case of decomposition of iron cobalt complex (IOC) showed

dominance of kinetic controlled mechanism based on random nucleation. The activation energy values calculated using four mechanism non-invoking equations for the dehydration and decomposition steps of all the three complex lies within the permissible limits. However, slight differences were observed, which are due to different methods based on different approaches and assumptions and so have different inherent limitations.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jaap.2006.03.006.

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